Raul Ababei

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Github: https://github.com/costitza

Gender: Male **Date of birth:** 13/07/2006 **Place of birth:** Romania **Nationality:** Romanian

ABOUT ME

Passionate about **robotics, programming, AI and ML**, I enjoy creating projects that combine **mechanics, electronics, and software**.
Motivated and hardworking, especially trying to learn new skills. Participated in **national and international competitions**, working on complex systems that strengthened my teamwork and problem-solving skills.
Experienced with **C++, Python, JavaScript, C#, PHP, React, Spring Boot, Flask, Node.js, .NET, PostgreSQL, Git, Pytorch, NextJS, Assembly** I love exploring new technologies and developing innovative solutions.
Currently studying **Computer Science at the University of Bucharest (FMI)**.

WORK EXPERIENCE

Preh Romania

City: Brasov | **Country:** Romania

[18/03/2024 – 22/03/2024] **Product development manager**

During a one-week internship at Preh Romania through the "Fit4Future" program, I explored the pre-production process and observed how engineering, design, and manufacturing teams work together to transform ideas into tangible products. The experience strengthened my passion for technology and the manufacturing industry.

CISPA European Cybersecurity & AI Championship https://cispa.de/en/cispa-hackathon-championship

City: Stockholm | **Country:** Sweden

[14/02/2026 – 15/02/2026] **Hackathon competitor**

- Selected to compete in a rigorous 24-hour pan-European hackathon focused on trustworthy AI and digital resilience at KTH Royal Institute of Technology.
- Collaborated in a 3-person team to solve difficult challenges at the intersection of machine learning and privacy.
- Developed solutions for three complex tasks: reconstructing private data via model inversion, analyzing ML model misclassification in image classification, and finding faulty data in label grouping (chimeras).

Kronbot - FIRST Tech Challenge Romania

City: Brasov | **Country:** Romania

[01/09/2022 – 01/06/2024] **Mechanical Team Leader**

- **Mechanical Design & Fabrication:** Directed the end-to-end design, rapid prototyping, and assembly of competitive robots, utilizing CAD modeling and custom fabrication techniques to build and optimize complex mechanisms.
- **Cross-Functional Leadership & Achievement:** Coordinated a specialized mechanical team and collaborated closely with software developers to integrate hardware and code, driving the team's successful qualification for the 2023 FTC Romania National Championship.
- **High-Pressure Troubleshooting:** Ensured continuous, reliable robot performance during intense competition environments by swiftly diagnosing mechanical failures and executing rapid, on-the-fly repairs.

EDUCATION AND TRAINING

[09/09/2021 – 06/06/2025] **Highschool Diploma**

Colegiul National de Informatica "Grigore Moisil"

City: Brasov | **Country:** Romania | **Field(s) of study:** Mathematics-Informatics | **Final grade:** 9.75

[30/09/2025 – Current] **Bachelor's Degree**

Universitatea din Bucuresti

City: Bucuresti | **Country:** Romania | **Field(s) of study:** Computer Science

- Coursework focused on core areas of computer science, including programming (C, C++, Python), algorithms and data structures, software engineering, databases, and computer networks. Additional studies covered embedded systems, artificial intelligence, and robotics, providing a strong foundation in both software development and hardware integration.

LANGUAGE SKILLS

Mother tongue(s): Romanian

Other language(s):

English

LISTENING C1 **READING** C1 **WRITING** B2

SPOKEN PRODUCTION B2 **SPOKEN INTERACTION** C1

German

LISTENING A2 **READING** A2 **WRITING** A2

SPOKEN PRODUCTION A1 **SPOKEN INTERACTION** A1

Levels: A1 and A2: Basic user; B1 and B2: Independent user; C1 and C2: Proficient user

PROJECTS

[01/05/2023 – 30/10/2023] **DpIT**

As part of this competition, I built a startup from scratch, bringing together my robotics experience and newly acquired business and entrepreneurship skills.

Link: https://www.kronsoft.ro/blog/gala-academiei-descopera-ti-pasiunea-in-it

[15/02/2024 – 01/08/2024] **Quest**

Designed and prototyped "**Quest**," a hybrid search-and-rescue robot capable of morphing between a four-wheeled rover and a quadcopter drone to navigate inaccessible environments. Engineered a multi-controller architecture using a **Raspberry Pi 4B** for high-level processing, an **Arduino Mega** for motor control, and a **Pixhawk 2.4.8** for flight stability, all powered by a custom power distribution circuit. Developed a custom 3D-printed PETG-CF chassis featuring a unique servo-actuated transformation mechanism and internal gear-driven wheels that allow for unobstructed flight. Implemented an autonomous object detection system using **YOLOv8n** on an **Oak-D Lite** camera, enabling the robot to identify and track victims in real-time while streaming low-latency video to a React-based web interface.

Participated in the **InfoEducație National Competition** and got **3rd** place, project developed by me and Motoasca Alexandru**.**

Link: https://github.com/costitza/Quest

[30/09/2021 – 01/08/2024] **InfoTron**

Over three consecutive editions of the **InfoTRON competition**, I developed multiple **robotics projects** that combined **mechanics, electronics, and programming** into fully functional systems. Among these, I designed multiple robots capable of autonomous driving and real-time environmental adaptability.

[30/11/2023 – 15/05/2024] **Maratonul de Educatie Antreprenoriala**

Participated in the **Entrepreneurship Marathon**, where I built a **startup concept and brand from scratch**. Our project focused on developing a **smart waste bin** capable of identifying and sorting recyclable materials automatically.

I contributed to creating the **business plan, branding, and product concept**, combining technical innovation with entrepreneurial thinking. This experience gave me insight into **how technology can address environmental challenges** while enhancing my teamwork, communication, and problem-solving skills.

[15/05/2025 – 29/07/2025] **uDraw**

Developed **uDraw**, a Windows-based **drawing application** built in **C# with Windows Forms** and connected to a **SQL Server database** for saving and loading drawings. Designed as a **personal learning project**, it focuses on combining UI development with database management.

Key features include creating and saving drawings to a database, loading previous sessions, and managing configurations through a custom settings file.

Technologies: C#, .NET Framework, Windows Forms, SQL Server

Link: https://github.com/costitza/udraw

[11/12/2025] **.flux**

Vibecoded **AI Engineering Context Dashboard** that secured 4th place at the ElevenLabs "AI Tinkerers" 4-hours Hackathon. I co-developed .flux with Tisca Catalin, an intelligent dashboard that unifies scattered engineering context. The system orchestrates specialized AI agents to synthesize data from GitHub repositories, PRs, and meeting transcripts, creating a semantic map of the codebase's evolution. This powers a multimodal voice assistant, enabling engineers to "talk" to their projects and retrieve real-time insights that traditional static tools overlook.

Link: https://github.com/costitza/dotflox

[13/03/2026 – Current] **Rippletrace**

Engineered a full-stack AI platform (Next.js, FastAPI) to predict and graphically visualize cascading global supply chain and stock market disruptions. Architected an automated 3-phase data pipeline ingesting live financial streams (Alpaca, Yahoo Finance) and dense SEC 10-K filings into a Neo4j Knowledge Graph, mapping complex multi-hop entity dependencies via Cypher. Orchestrated Graph Retrieval-Augmented Generation (GraphRAG) using LangChain and Groq-hosted Llama 3 models (8B/70B) for real-time NLP extraction, enabling hallucination-free risk assessments and multi-tier predictive market simulations. Technologies: Next.js, FastAPI, Python, Neo4j, Cypher, LangChain, Groq, Llama 3

Link: https://github.com/costitza/rippletrace

SKILLS

Microsoft Office | Microsoft Word | Microsoft Powerpoint | C++ | Python | JavaScript | PHP | C# | SQL | HTML | CSS | Git | Critical Analysis and Problem Solving | Software Develoment | Robot programming | Active learning | AI | Agentic workflows | JSON | Linux | ML | Pytorch | NextJS

HONOURS AND AWARDS

[05/08/2024] **InfoEducatie National Olympiad - 3rd place Awarding institution:** Ministerul Educatiei

[10/03/2023] **First Tech Challenge National Qualification Awarding institution:** FTC Romania

[10/05/2022] **Infotron 2nd edition - 1st place Awarding institution:** Informatica, pasul tau spre lumea mecatronicii - Infotron_2022

[15/06/2024] **Infotron 4th edition - 1st place Awarding institution:** Informatica, pasul tau spre lumea mecatronicii - Infotron_2024

[27/10/2024] **Robochallenge 15th edition - freestyle showcase - honorable mention Awarding institution:** R obochallenge

[10/03/2023] **Informatics Regional Olympiad - Honorable mention Awarding institution:** Ministerul Educatiei